

3D Scatter Plot Projection

Introduction

The project demonstrates visualization of randomly generated 3D points on an HTML5 canvas using JavaScript. It allows users to rotate the point cloud and compare Orthographic and Perspective projection techniques.

Project Overview

This application loads 3D coordinates, applies mathematical transformations, and displays them interactively with rotation controls and animation.

Objectives

- Visualize 3D points.
- Apply Yaw, Pitch and Roll rotations.
- Demonstrate Orthographic and Perspective projections.
- Provide interactive controls and smooth animation.

Workflow

1. Generate random 3D points.
2. Store them in data.json.
3. Load data into JavaScript.
4. Convert degrees to radians.
5. Apply Pitch, Yaw and Roll rotations.
6. Translate points along the Z-axis.
7. Apply Orthographic or Perspective projection.
8. Draw the points on the canvas.
9. Continuously redraw using `requestAnimationFrame()`.

Random Number Generation

Random x, y and z coordinates are generated within a fixed range (approximately -400 to +400) to create a 3D scatter cloud.

Rotation Formulas

Pitch: $y' = y \cos \theta - z \sin \theta$, $z' = y \sin \theta + z \cos \theta$

Yaw: $x' = x \cos \theta + z \sin \theta$, $z' = -x \sin \theta + z \cos \theta$

Roll: $x' = x \cos \theta - y \sin \theta$, $y' = x \sin \theta + y \cos \theta$

Degree to Radian Conversion

Radians = Degrees $\times \pi / 180$. JavaScript trigonometric functions require angles in radians.

Orthographic Projection

Formula: $x' = x$, $y' = y$. Depth is ignored and object size remains constant.

Perspective Projection

Scale = $f / (f + z)$, $x' = x \times \text{Scale}$, $y' = y \times \text{Scale}$. Distant objects appear smaller, giving a realistic 3D effect.

Overall Implementation

The dataset is loaded, rotations are applied, projection is selected, and the resulting 2D coordinates are rendered on the canvas. Auto Rotate continuously updates the scene.

Advantages

Simple, interactive, lightweight, educational, and useful for learning computer graphics concepts.

Applications

Computer graphics, scientific visualization, engineering, education, game development, and data visualization.

Conclusion

The 3D Scatter Plot Projection project successfully demonstrates the fundamentals of 3D graphics using HTML5 Canvas and JavaScript. It combines mathematical transformations, projection techniques, and animation to create an interactive visualization. The project helps users understand how 3D objects are represented on a 2D screen while providing a practical introduction to computer graphics concepts.